



Reshoring, automation, and labor markets under trade uncertainty^{☆,☆☆,★}

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ABSTRACT

We study the implications of trade uncertainty for reshoring, automation, and U.S. labor markets. Rising trade uncertainty creates incentives for firms to reduce exposure to foreign suppliers by moving production and distribution processes to domestic producers. However, we argue that reshoring does not necessarily bring jobs back to the home country or boost domestic wages, especially when firms have access to labor-substituting technologies such as automation. Automation improves labor productivity and facilitates reshoring, but it can also displace jobs. Furthermore, automation poses a threat that weakens the bargaining power of unskilled workers in wage negotiations, depressing their wages and raising the skill premium and wage inequality. Our model predictions are in line with industry-level empirical evidence.

1. Introduction

The COVID-19 pandemic has exposed important vulnerabilities in global supply chains. Ongoing trade tensions as well as increasing risks from climate change and geopolitical conflicts are making global production strategies riskier than in the past.

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In this new economic environment, moving some production and distribution processes from abroad back to domestic suppliers (i.e., reshoring) is becoming an increasingly attractive option to mitigate the risks of supply chain disruptions.¹

How this process will unfold and what the impacts on labor markets will be remain highly uncertain. One possibility is that reshoring could increase jobs in the home country and boost wages for domestic workers, reversing the effects of the China shock originally studied by Autor et al. (2013). In this paper, we argue that reshoring may not necessarily increase domestic employment and wages when labor-substituting technologies, such as automation, are available for firms to lower production costs.

Over the past three decades, advanced economies that offshored production processes have also experienced a steady increase in the adoption of automation technologies, such as artificial intelligence, machine learning, and robotics. Empirical evidence suggests that automation raises labor productivity (Graetz and Michaels, 2018) and reduces unit labor costs and worker wages (Acemoglu and Restrepo, 2020). The increased ability to automate labor-intensive production processes could reduce firms' need to offshore production to contain labor costs. In line with these changing incentives, import growth has slowed significantly relative to GDP since the trade collapse during the Great Recession.

Coupled with a greater ability to automate, recent increases in trade uncertainty may have accelerated the trend in reshoring. While reshoring tends to raise domestic labor demand and real wages, firms' options to automate help mitigate the increase in labor costs, since it acts as a threat against workers – especially unskilled workers who can be easily substituted by robots – in wage bargaining. This automation threat channel – originally studied by Leduc and Liu (2024) – helps contain the rise in labor costs, reinforcing the incentive for reshoring. Since robots substitute for unskilled workers and complement skilled workers, increased automation spurred by reshoring may also raise the skill premium and income inequality.

In this paper, we formalize this perspective by developing a macro framework featuring automation, heterogeneous worker skills, and international trade frictions. We use this framework to examine the impacts of a rise in trade uncertainty on reshoring, automation, and domestic labor markets. We generalize the automation threat channel to a small open economy with trade in intermediate inputs. Trade is subject to time-varying iceberg costs with stochastic volatility meant to capture trade uncertainty arising from geopolitical, climate, and trade policy risks. To produce a final good, firms use a mixture of domestic and foreign intermediate goods. We capture the interaction between reshoring and automation by assuming that domestic intermediate goods producers can use two types of technologies: a labor-only technology that uses unskilled workers and an automated process that uses both robots and skilled workers as inputs.²

We assume that unskilled workers search for jobs in a frictional labor market, subject to search frictions as in the standard Diamond-Mortensen-Pisarides (DMP) framework. Unskilled wages are determined by Nash bargaining between a firm and a worker. Because firms have the option to automate unfilled vacancies, the threat of automation acts as an outside option for the firm and weighs on bargained wages.³ This effect is compounded when firms do actually automate, since the associated productivity boost lowers domestic marginal costs of production further.

In our framework, heightened trade uncertainty operates through three key channels. First, trade uncertainty has an expenditure-switching effect that redirects the demand for intermediate goods toward domestic producers (i.e., reshoring).⁴ This expenditure-switching effect stimulates automation investment, raising the demand for skilled workers. While the expenditure-switching effect and a greater use of automated processes have a job-creating effect through raising the value of unfilled vacancies, this channel is more than offset by the job-displacing effect of automation on unskilled workers. To elaborate, in a frictional labor market with long-term employment relations, firms respond to heightened uncertainty by reducing hiring and increasing automation to meet the increased demand for domestic goods. Second, trade uncertainty also generates greater precautionary savings, which reduces the real interest rate and further stimulates automation. Third, heightened trade uncertainty raises the option value of waiting, discouraging automation investment.

We show that, with our calibration, the positive effects from expenditure switching and precautionary savings dominate the negative option-value effect, such that trade uncertainty boosts automation, raises unemployment for unskilled workers, and also raises the skill premium. These effects of trade uncertainty are amplified for an economy that is more open to trade, has more automated production, or faces more persistent trade uncertainty.

Our model produces a rich set of empirically testable predictions. First, the model predicts that an increase in trade uncertainty increases reshoring and stimulate automation investment. Second, increased automation triggered by trade uncertainty raises labor productivity and value added. Third, the threat of automation depresses wages and employment of unskilled workers while raising wages of skilled workers, resulting in an increase in the skill premium. These effects should be stronger in an economy more open to international trade.

¹ According to a Thomas Industrial Survey, about two-thirds of North American manufacturers reported they are likely to bring manufacturing production and sourcing back to North America because of concerns about the global supply chain disruptions following the COVID-19 pandemic. In addition, about a quarter of those manufacturers are considering expanding industrial automation.

² We focus on automation decisions at the business cycle frequency. However, automation can also be the result of long-run technological improvements that can allow the automation of tasks previously done by labor. We view this form of automation as occurring relatively infrequently and instead focus on an environment with fixed production technologies.

³ Unlike the standard DMP framework, we assume that vacancy creation incurs a random fixed cost (Fujita and Ramey, 2007), such that an unfilled vacancy retains value in equilibrium and captures the firms' outside option and ability to automate in the future.

⁴ To keep the analysis tractable, we model reshoring or offshoring in a reduced-form way. We do not model firms' choices of production locations. We interpret importing of intermediate goods as production that could have been done domestically but is instead *offshored*. Similarly, we interpret a decline in imports of intermediate goods as *reshoring*.

The model predictions are consistent with empirical evidence from industry-level data. We use data on industrial robots, intermediate goods imports, employment, value-added, and wages in two-digit International Standard Industrial Classification (ISIC, Rev. 4) industries from 1997 to 2022 to construct measures of automation, offshoring, labor productivity, and the skill premium.⁵ We measure trade uncertainty using aggregate trade policy uncertainty (TPU) constructed by [Caldara et al. \(2020\)](#), interacted with a measure of initial exposure to offshored production. We show that, controlling for industry and time fixed effects, an increase in trade uncertainty is associated with larger increases in automation and larger declines in offshoring in industries that are more exposed to offshoring.⁶

We also find that an increase in trade uncertainty is associated with larger increases in labor productivity and value added in industries that are more exposed to offshoring and that these effects work partly through an automation channel. We examine the channeling effects using a two-stage least squares approach ([Bertrand and Mullainathan, 2001](#)). In the first stage, we regress a measure of automation (robot density) on trade uncertainty, controlling for industry and time fixed effects. In the second stage, we regress each variable of interest (including labor productivity, employment, value-added, and the skill premium) on robot density predicted from the first-stage regression. The estimated coefficient in the second-stage regression indicates the sensitivity of each of the macroeconomic variables to changes in robot density that comes from trade policy uncertainty. We find that an increase in robot density driven by trade uncertainty is associated with an increase in labor productivity, value added, and the skill premium.⁷

Our work contributes to a relatively new but growing literature on the effects of reshoring. Empirically, drawing clear conclusions about the effects of reshoring has been challenging given the novelty of the practice and thus the lack of data. Nonetheless, a few papers have assessed the empirical links between reshoring and automation. For instance, [Dachs et al. \(2019\)](#) find a positive relationship between reshoring and investment in Industry 4.0 technologies for 1700 firms in Austria, Germany, and Switzerland. More broadly, our paper is also related to the literature on the effects of trade policy on the structure of trade and global supply chains ([Fajgelbaum et al., 2021](#); [Alfaro and Chor, 2023](#); [Utar et al., 2023](#); [Grossman et al., 2024](#)).⁸

By emphasizing the effects of uncertainty on reshoring and automation, our paper complements recent work that examines the effects of changes in automation on trade. In particular, a growing body of literature has documented the interaction between automation and offshoring and showed that automation tends to reduce offshoring ([De Backer et al., 2018](#); [Artuc et al., 2019](#); [Stemmler, 2019](#); [Faber, 2020](#); [Carbonero et al., 2020](#); [Krenz et al., 2021](#); [Bonfiglioli et al., 2022](#)).⁹ [Mandelman and Zlate \(2022\)](#) argue that offshoring and automation reduce employment and wages of middle-skill occupations but enhance those for high-skilled ones. We examine the nexus between offshoring and automation from a different angle by showing how trade uncertainty induces reshoring and boosts automation investment and how the interactions between reshoring and automation affect the responses of domestic labor market variables to trade uncertainty.

Our paper also adds to an extensive literature on the effects of trade policy uncertainty (e.g., [Handley and Limão, 2015, 2017, 2022](#); [Feng et al., 2017](#); [Crowley et al., 2018](#); [Alessandria et al., 2019, 2021](#); [Poilly and Tripier, 2023](#); [Choi et al., 2023](#); [Correa et al., 2023](#); [Alessandria et al., 2024](#); [Rodrigue et al., 2024](#)), and more broadly, on the macroeconomic effects of uncertainty (e.g., [Bloom, 2009](#); [Fernández-Villaverde et al., 2011b](#); [Alessandria et al., 2015](#); [Leduc and Liu, 2016](#); [Basu and Bundick, 2017](#); [Greenland et al., 2019](#); [Dur et al., 2024](#); [Kim and Lee, 2024](#)). Related to our study, [Novy and Taylor \(2020\)](#) argue that trade flows can be more sensitive to uncertainty shocks than domestic production because of higher fixed costs of orders of foreign inputs. [Caldara et al. \(2020\)](#) show that an increase in TPU reduces business investment, both in the data and in an open-economy model. [Faber et al. \(2023\)](#) empirically show that country-level uncertainty in the developing world induces reshoring to developed countries, but only if automation technology is available. Complementary to these studies, our paper highlights how trade uncertainty can drive three-way interactions between reshoring, automation, and labor markets.

2. The model

This section presents a small open economy model featuring labor search frictions, endogenous decisions of automation, and offshoring.

⁵ Due to data limitations, this paper focuses on industrial robots, a specific type of automation technology. We use the terms “automation” and “robots” interchangeably. Robotics constitute an important component of the automation technology. According to the 2019 Annual Business Survey, 8.7% of U.S. manufacturing firms, which account for 45% of manufacturing employment, utilize robotics in their production processes ([Acemoglu et al., 2022](#)).

⁶ Our model's prediction that trade uncertainty raises automation investment does not necessarily contradict the empirical finding of [Caldara et al. \(2020\)](#) that trade uncertainty reduces business investment (such as nonresidential structures and general capital equipment). It would be straightforward to generalize our model to incorporate business investment. For example, one could modify the traditional technology (i.e., the non-automation technology) that uses unskilled labor as the only input in our baseline model by assuming that both capital and labor are required as input factors. We conjecture that, in such a model, an increase in trade uncertainty could boost automation investment, which in turn could displace unskilled jobs and reduce business investment, in line with the findings of [Caldara et al. \(2020\)](#).

⁷ While these results are broadly in line with our theoretical predictions, we note that we are using a relatively small sample of industries over a relatively short time period and therefore one should interpret our empirical results with caution.

⁸ The literature has also studied the importance of global supply chains in optimal trade policy; see, for example, [Blanchard et al. \(2017\)](#), [Grossman et al. \(2023\)](#) and [Antràs et al. \(2024\)](#).

⁹ [Artuc et al. \(2023\)](#) show that, by increasing productivity, robotization in the North increases imports from the South. [Baur et al. \(2023\)](#) document that the impact of automation on trade depends on input–output linkages.

2.1. Key features in the model

Final consumption goods are produced using intermediate goods that are imported or domestically produced. Domestic intermediate goods can be produced using two types of technologies, a labor-only technology that uses unskilled workers as the only input and an automation technology that uses both robots and skilled workers as inputs.

We assume that a firm that chooses to use the automation technology can adopt a robot at a random sunk cost and hire a skilled worker from a competitive spot skilled labor market. If the firm chooses to operate the labor-only technology, then it can hire an unskilled worker subject to labor market search frictions in the spirit of the standard DMP framework.

In the beginning of a period t , firms carry over the stock of unfilled vacancies from the previous period, a fraction of which is automated by adopting robots. The stock of vacancies v_t available for hiring workers consists of the remaining vacancies after automation, the jobs separated in the beginning of the period, and newly created vacancies. The job seekers (the mass of which is u_t) randomly match with the vacancies (v_t) in the labor market, with the number of new matches (m_t) determined by a matching technology. Production then takes place, using either a labor-only or an automation technology. The unfilled vacancies and the pool of employed workers at the end of the period are carried over to the next period, and the same sequence of economic activities repeats in period $t + 1$.

Compared to the standard DMP model, our model introduces four new features. First, we replace the free-entry assumption in the DMP model with costly vacancy creation, as in Fujita and Ramey (2007) and Leduc and Liu (2020). Since creating a new vacancy incurs a fixed cost, a vacancy has a positive value even if it is not filled by an unskilled worker. The number of vacancies becomes a slow-moving state variable (instead of a jump variable, as in the standard DMP framework), enabling our model to match the persistent vacancy dynamics in the data.

Second, we introduce endogenous automation decisions. In the beginning of period t , each firm draws a sunk cost of automation, which determines whether the firm will automate production or post the vacancy for hiring a worker. If the automation cost lies below a threshold value, then the firm automates production by adopting a robot and hiring skilled workers to operate the robot. In that case, the firm obtains the automation value and the vacancy would be taken offline. If the automation cost exceeds the threshold, then the firm posts the vacancy for hiring an unskilled worker.

Third, we allow for worker skill heterogeneity, with skilled and unskilled workers, who are all members of the representative household. In our model, robots and skilled workers are complementary inputs, whereas they are substitutes for unskilled workers. This feature allows us to examine the joint effects of automation and offshoring on employment of workers with different skills and also on income inequality stemming from the skill premium.

Fourth, we introduce offshoring by allowing final goods producers to import intermediate goods. Changes in trade costs caused by, for example, global supply chain disruptions or trade wars can affect the effective costs of offshoring, which in turn affects the relative demand for intermediate goods that are imported versus domestically produced. Such changes in relative demand in turn drive changes in automation decisions, employment, and income distribution.

2.2. The frictional labor market for unskilled workers

At the beginning of period t , there are N_{t-1} existing job matches for unskilled workers. The measure of unemployed job seekers is given by

$$u_t = 1 - (1 - \delta)N_{t-1}, \quad (1)$$

where $\delta \in (0, 1)$ denotes the job separation rate and we have assumed full labor force participation with the size of unskilled labor normalized to one.

The stock of vacancies v_t at the beginning of period t consists of unfilled vacancies carried over from period $t - 1$ that are not automated plus the separated employment matches and newly created vacancies. The law of motion for vacancies is given by

$$v_t = (1 - q_{t-1}^v)(1 - q_t^a)v_{t-1} + \delta N_{t-1} + \eta_t, \quad (2)$$

where q_{t-1}^v denotes the job filling rate in period $t - 1$, q_t^a denotes the automation probability in period t , and η_t denotes newly created vacancies (i.e., entry).

In the labor market, new job matches (denoted by m_t) are formed between job seekers and open vacancies based on the matching function

$$m_t = \mu u_t^\alpha v_t^{1-\alpha}, \quad (3)$$

where μ is a scale parameter that measures matching efficiency and $\alpha \in (0, 1)$ is the elasticity of job matches with respect to the number of job seekers.

The flow of new job matches adds to the employment pool, whereas job separations subtract from it. Aggregate employment evolves according to the law of motion

$$N_t = (1 - \delta)N_{t-1} + m_t. \quad (4)$$

At the end of period t , the searching workers who failed to find a job remain unemployed. Thus, unemployment is given by

$$U_t = u_t - m_t = 1 - N_t. \quad (5)$$

For convenience, we define the job finding probability q_t^u as

$$q_t^u = \frac{m_t}{u_t}. \quad (6)$$

Similarly, we define the vacancy filling probability q_t^v as

$$q_t^v = \frac{m_t}{v_t}. \quad (7)$$

2.3. The representative household

The representative household has the utility function

$$\mathbb{E} \sum_{t=0}^{\infty} \beta^t (\ln C_t - \chi N_t), \quad (8)$$

where $\mathbb{E}[\cdot]$ is an expectation operator, $\beta \in (0, 1)$ is a subjective discount factor, C_t denotes consumption, and N_t denotes the fraction of unskilled household members who are employed.

The representative household faces the sequence of budget constraints

$$C_t + B_t = r_{t-1} B_{t-1} + w_{nt} N_t + w_{st} \bar{s} + \phi(1 - N_t) + d_t - T_t, \quad \forall t \geq 0, \quad (9)$$

where B_t denotes the household's holdings of a risk-free bond (in units of final goods) at the real interest rate r_t ; w_{nt} and w_{st} denote the real wage rates of unskilled and skilled workers (also in units of final consumption goods), respectively; d_t denotes the household's share of firm profits; and T_t denotes lump-sum taxes. The parameter ϕ measures the flow benefits of unemployment. For simplicity, we assume that the aggregate supply of skilled labor is fixed at $\bar{s}$.

Denote by $V_t(B_{t-1}, N_{t-1})$ the value function for the representative household. The household's optimizing problem can be written in the recursive form

$$V_t(B_{t-1}, N_{t-1}) \equiv \max_{C_t, N_t, B_t} \ln C_t - \chi N_t + \mathbb{E}_t D_{t,t+1} V_{t+1}(B_t, N_t), \quad (10)$$

subject to the budget constraint (9) and the employment law of motion (4) for unskilled workers, which can be written as

$$N_t = (1 - \delta)N_{t-1} + q_t^u u_t, \quad (11)$$

where we have used the definition of the job finding probability q_t^u with the measure of job seekers u_t . In the optimizing decisions, the household takes the economy-wide job finding rate q_t^u as given.

The stochastic discount factor (SDF) is given by

$$D_{t,t+1} \equiv \beta \frac{\Lambda_{t+1}}{\Lambda_t}, \quad (12)$$

where Λ_t denotes the Lagrange multiplier for the budget constraint (9).

We define the employment surplus (i.e., the value of employment relative to unemployment) as $S_t^H \equiv \frac{1}{\Lambda_t} \frac{\partial V_t(B_{t-1}, N_{t-1})}{\partial N_t}$. The optimizing decision for employment implies that the employment surplus satisfies the Bellman equation

$$S_t^H = w_{nt} - \phi - \frac{\chi}{\Lambda_t} + \mathbb{E}_t D_{t,t+1} (1 - q_{t+1}^u)(1 - \delta) S_{t+1}^H. \quad (13)$$

The employment surplus has a straightforward economic interpretation. If the household adds a new unskilled worker in period t , then the current-period gain would be wage income net of the opportunity costs of working, including unemployment benefits and the disutility of working. The household also enjoys the continuation value of employment if the employment relation continues. Having an extra unskilled worker today adds to the employment pool tomorrow (provided that the employment relation survives job separation); however, adding a worker today would also reduce the pool of searching workers tomorrow, a fraction q_{t+1}^u of whom would be able to find jobs. Thus, the marginal effect of adding a new worker in period t on employment in period $t+1$ is given by $(1 - q_{t+1}^u)(1 - \delta)$, resulting in the effective continuation value of employment shown in the last term of Eq. (13).

Finally, the household's optimizing consumption-savings decision implies the intertemporal Euler equation

$$1 = \mathbb{E}_t D_{t,t+1} r_t. \quad (14)$$

2.4. Final goods production

A homogeneous final good is produced using two types of intermediate inputs, one produced by domestic firms (denoted by Y_{dt}) and the other imported from the foreign country (Y_{ft}). Importing goods is subject to a delivery lag such that imported intermediate goods today can be used for final goods production tomorrow.¹⁰ The production function of final goods is given by

$$Y_t = \left[\alpha_d^{\frac{1}{\theta}} Y_{dt}^{\frac{\theta-1}{\theta}} + (1 - \alpha_d)^{\frac{1}{\theta}} Y_{f,t-1}^{\frac{\theta-1}{\theta}} \right]^{\frac{\theta}{\theta-1}}, \quad (15)$$

¹⁰ We incorporate delivery lags for imported inputs to enhance the realism of our model. As demonstrated in Appendix D, the results remain qualitatively similar even when these delivery lags are excluded.

where the parameter θ measures the elasticity of substitution between home-produced and imported intermediate goods, and the parameter α_d measures the importance of domestic intermediate goods for final goods production. We assume that intermediate goods are tradable while final goods are nontradable. To keep the analysis tractable, we interpret importing of intermediate goods as part of the production that could have been undertaken domestically but is instead offshored.¹¹

We denote by p_{dt} and p_{ft} the relative prices of intermediate goods (i.e., in units of final consumption goods) produced domestically and imported, respectively. The relative price of imported goods faced by domestic final goods producers is given by

$$p_{ft} = \frac{\tau_t P_t^*}{P_t} = \tau_t Q_t, \quad (16)$$

where τ_t denotes an iceberg trade cost, P_t is the price of final consumption goods, P_t^* is the foreign price level, and $Q_t \equiv \frac{P_t^*}{P_t}$ is the real exchange rate (RER). The small open economy takes the foreign price level P_t^* as exogenously given. Without loss of generality, we normalize $P_t^* = 1$ such that the real exchange rate is isomorphic to the domestic price level.

We assume that, for every unit of goods delivered to the destination, $\tau_t > 1$ units of goods need to be shipped. The trade cost τ_t is an exogenous process with a time-varying volatility, which captures trade uncertainty related to factors such as trade wars, geopolitical tensions, or climate change risks that might cause global supply chain disruptions. Specifically, we assume that the trade cost follows the stationary stochastic process

$$\ln(\tau_t) = (1 - \rho_\tau) \ln(\bar{\tau}) + \rho_\tau \ln(\tau_{t-1}) + \sigma_{\tau t} \varepsilon_{\tau t}, \quad (17)$$

where $\bar{\tau}$ is the mean of τ_t , $\rho_\tau \in (-1, 1)$ is a persistence parameter, and $\varepsilon_{\tau t}$ is a white noise innovation. The term $\sigma_{\tau t}$ is a stochastic volatility of the trade cost shock, which we interpret as trade uncertainty, and it follows the process

$$\sigma_{\tau t} = (1 - \rho_{\sigma\tau}) \sigma_\tau + \rho_{\sigma\tau} \sigma_{\tau,t-1} + \eta_\tau u_{\tau t}. \quad (18)$$

Here, $\rho_{\sigma\tau} \in (-1, 1)$ is the persistence and η_τ is the standard deviation of the trade uncertainty shock, $u_{\tau t}$ is a white noise innovation, and σ_τ is the average standard deviation of the trade cost shock.¹²

Final goods producers take all prices as given and choose Y_{dt} and Y_{ft} to maximize the expected present value of profit flows. The optimizing problem is described by the Bellman equation

$$V_t(Y_{f,t-1}) = \max_{Y_{dt}, Y_{ft}} Y_t - p_{dt} Y_{dt} - p_{ft} Y_{ft} + \mathbb{E}_t D_{t,t+1} V_{t+1}(Y_{ft}), \quad (19)$$

subject to the technology constraint (15), where $V_t(Y_{f,t-1})$ denotes the value function, which depends on the state variable $Y_{f,t-1}$. The first-order conditions for this optimizing problem are given by

$$p_{dt} = \frac{\partial Y_t}{\partial Y_{dt}}, \quad p_{ft} = \mathbb{E}_t D_{t,t+1} V'_{t+1}(Y_{ft}). \quad (20)$$

The envelope condition implies that

$$V'_t(Y_{f,t-1}) = \frac{\partial Y_t}{\partial Y_{f,t-1}}. \quad (21)$$

Combining (20) and (21), we obtain

$$p_{dt} = \left(\frac{\alpha_d Y_t}{Y_{dt}} \right)^{\frac{1}{\theta}}, \quad p_{ft} = \mathbb{E}_t D_{t,t+1} \left(\frac{(1 - \alpha_d) Y_{t+1}}{Y_{ft}} \right)^{\frac{1}{\theta}}. \quad (22)$$

The domestic intermediate good is itself a Constant Elasticity of Substitution (CES) aggregate of two types of intermediate goods produced using labor-only technology and automation technology. In particular, the quantity of domestically produced intermediate goods Q_{dt} is given by

$$Q_{dt} = \left[\alpha_n^{\frac{1}{\sigma}} Y_{nt}^{\frac{\sigma-1}{\sigma}} + (1 - \alpha_n)^{\frac{1}{\sigma}} Y_{at}^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}}, \quad (23)$$

where Y_{nt} denotes the intermediate goods produced using the labor-only technology, Y_{at} denotes the intermediate goods produced using the automation technology, the parameter σ is the elasticity of substitution between the two types of intermediate goods, and the parameter α_n governs the relative importance of Y_{nt} in the aggregation technology.

Some domestically produced intermediate goods are exported to the foreign country. Thus, we have

$$Q_{dt} = Y_{dt} + \tau_t X_t, \quad (24)$$

where X_t denotes the quantity of exports.

¹¹ In addition, we treat the rest of the world as a uniform area subject to the same degree of trade uncertainty. Thus, we abstract from the possibility that higher trade uncertainty in a specific region could lead firms to diversify the sourcing of their products to other regions. While that is an interesting and relevant issue, it is beyond the scope of this paper.

¹² Appendix D.2 shows that the effects of a second-moment shock to the tariff rate are similar to those of a second-moment shock to the iceberg costs.

The optimal choices of domestic intermediate goods producers imply that

$$\frac{p_{nt}}{p_{dt}} = \left(\frac{\alpha_n Y_{dt}}{Y_{nt}} \right)^{\frac{1}{\sigma}}, \quad \frac{p_{at}}{p_{dt}} = \left(\frac{(1 - \alpha_n) Y_{dt}}{Y_{at}} \right)^{\frac{1}{\sigma}}. \quad (25)$$

The zero-profit condition for domestic intermediate goods producers implies that

$$p_{dt} = \left[\alpha_n p_{nt}^{1-\sigma} + (1 - \alpha_n) p_{at}^{1-\sigma} \right]^{\frac{1}{1-\sigma}}. \quad (26)$$

Discussion. To maintain tractability, our model abstracts from a few potentially important adjustment channels. On the household side, we exclude potential effects of automation on labor force participation (e.g., Grigoli et al., 2020) or the reallocation of workers to the services sector (Autor and Dorn, 2013). However, these adjustment channels are more likely to be quantitatively important in the longer term rather than at the business cycle frequency, which is the focus of our paper. Taking these additional adjustment channels into account would likely reduce the impact of trade uncertainty on unemployment in the long run. On the production side, we abstract from firms' ability to smooth unexpected changes in demand arising from trade uncertainty through variations in inventories (e.g., Alessandria et al., 2019).

2.5. Domestic production of intermediate goods

A firm makes automation decisions at the beginning of the period t . Adopting a robot requires a sunk cost v in units of consumption goods, which is drawn from the *i.i.d.* distribution $G(v)$.¹³ A firm chooses to adopt a robot if and only if the cost of automation is less than the benefit. For any given benefit of automation, there exists a threshold value v_t^* in the support of the distribution $G(v)$, such that automation occurs if and only if $v \leq v_t^*$. If the firm adopts a robot to replace the job position, then the vacancy will be taken offline and will not be available for hiring a worker. Thus, the automation threshold v_t^* depends on the value of automation (denoted by J_t^a) relative to the value of a vacancy (denoted by J_t^v). In particular, the threshold for automation decision is given by

$$v_t^* = J_t^a - J_t^v. \quad (27)$$

The probability of automation is then given by the cumulative density of the automation costs evaluated at v_t^* . That is,

$$q_t^a = G(v_t^*). \quad (28)$$

The flow of automated job positions adds to the stock of automated positions (denoted by A_t), which becomes obsolete at the rate $\rho^o \in [0, 1]$ in each period. Thus, A_t evolves according to the law of motion

$$A_t = (1 - \rho^o) A_{t-1} + q_t^a (1 - q_{t-1}^v) v_{t-1}, \quad (29)$$

where $q_t^a (1 - q_{t-1}^v) v_{t-1}$ is the number of newly automated job positions.¹⁴

If the firm adopts a robot, then it optimally chooses the input of skilled workers s_t , with the production function

$$y_{at} = Z_t \zeta^{\gamma_a} s_t^{1-\gamma_a}, \quad (30)$$

where $\gamma_a \in (0, 1)$ denotes the elasticity of output with respect to the robot input, Z_t denotes a total factor productivity (TFP) shock, and ζ denotes an automation-specific productivity.¹⁵

TFP follows a stationary $AR(1)$ stochastic process

$$\ln(Z_t) = (1 - \rho_z) \ln(\bar{Z}) + \rho_z \ln(Z_{t-1}) + \sigma_z \varepsilon_{zt}, \quad (31)$$

where $\bar{Z}$ is the mean of Z_t , $\rho_z \in (-1, 1)$ is a persistence parameter, ε_{zt} is a white noise innovation, and σ_z is the standard deviation of the TFP shock.¹⁶

The firm takes the skilled real wage rate w_{st} as given and chooses s_t to maximize the profit before paying the robot operation cost κ_a . The value of automation is then given by

$$J_t^a = \pi_t^a (1 - \kappa_a) + (1 - \rho^o) \mathbb{E}_t D_{t,t+1} J_{t+1}^a, \quad (32)$$

¹³ The assumption that adopting a robot requires a sunk cost implies that the automation decision is irreversible. This irreversibility tends to reduce the incentive to automate in response to trade uncertainty, making our results conservative. In contrast, if the automation decision was reversible (e.g., involving a per-period fixed cost of automation), the automation response to trade uncertainty would likely be stronger, as the option value of delaying automation would no longer exist in this case.

¹⁴ If a vacancy is "filled" by a robot, it will be taken offline once and for all. Even if the robot later becomes obsolete, the vacated position does not return to the stock of vacancies.

¹⁵ In our baseline model, the use of robots is not subject to trade costs. However, in practice, firms' automation technology may partly be imported and thus subject to trade costs and trade uncertainty. We consider this more general case in a robustness exercise below.

¹⁶ We focus on trade uncertainty in the main analysis, although we also examine the effects of TFP uncertainty, which is measured by time-varying volatility of the TFP shock (see Appendix D).

where $\pi_t^a \equiv \max_{s_t} p_{at} Z_t \zeta^{\gamma_a} s_t^{1-\gamma_a} - w_{st} s_t = \gamma_a p_{at} Z_t \zeta^{\gamma_a} s_t^{1-\gamma_a}$.

If the automation sunk cost exceeds the threshold v_t^* , then the firm chooses not to adopt a robot and instead chooses to post the vacancy in the labor market for hiring an unskilled worker. In addition, newly separated jobs and newly created vacancies add to the stock of vacancies for hiring unskilled workers. We assume that creating a new vacancy incurs an entry cost e in units of consumption goods, which is drawn from an *i.i.d.* distribution $F(e)$. A new vacancy is created if and only if the net value of entry is non-negative. The benefit of creating a new vacancy is the vacancy value J_t^v . Thus, the number of new vacancies η_t is given by the cumulative density of the entry costs evaluated at J_t^v . That is,

$$\eta_t = F(J_t^v). \quad (33)$$

Posting a vacancy incurs a per-period fixed cost κ (in units of final consumption goods). If the vacancy is filled (with probability q_t^v), the firm obtains the employment value J_t^e . Otherwise, the firm carries over the unfilled vacancy to the next period, which will be automated with the probability q_{t+1}^a . If the vacancy is automated, then the firm obtains the automation value J_{t+1}^a net of the expected robot adoption costs; otherwise, the vacancy will remain open, and the firm receives the vacancy value J_{t+1}^v . Thus, the vacancy value satisfies the Bellman equation

$$J_t^v = -\kappa + q_t^v J_t^e + (1 - q_t^v) \mathbb{E}_t D_{t,t+1} \left\{ q_{t+1}^a J_{t+1}^a - \int_0^{v_{t+1}^*} v dG(v) + (1 - q_{t+1}^a) J_{t+1}^v \right\}. \quad (34)$$

If a firm successfully hires an unskilled worker, then it can produce Z_t units of intermediate goods. The value of employment satisfies the Bellman equation

$$J_t^e = p_{nt} Z_t - w_{nt} + \mathbb{E}_t D_{t,t+1} \{ (1 - \delta) J_{t+1}^e + \delta J_{t+1}^v \}. \quad (35)$$

Hiring a worker generates a flow profit $p_{nt} Z_t - w_{nt}$ in the current period (in final consumption units). If the job is separated in the next period (with probability δ), then the firm receives the vacancy value J_{t+1}^v . Otherwise, the firm receives the continuation value of employment.

2.6. The Nash bargaining wage

When a job match is formed, the wage rate is determined through Nash bargaining. The bargaining wage splits the joint surplus of a job match between the unskilled worker and the firm. The worker's employment surplus is given by S_t^H in Eq. (13). The firm's surplus is given by $J_t^e - J_t^v$. The possibility of automation affects the value of a vacancy and thus indirectly affects the firm's reservation value and its bargaining decisions.

The Nash bargaining problem is given by

$$\max_{w_{nt}} (S_t^H)^b (J_t^e - J_t^v)^{1-b}, \quad (36)$$

where $b \in (0, 1)$ represents the bargaining weight for workers.

Define the total surplus as

$$S_t \equiv J_t^e - J_t^v + S_t^H. \quad (37)$$

Then the Nash bargaining solution is given by

$$J_t^e - J_t^v = (1 - b) S_t, \quad S_t^H = b S_t. \quad (38)$$

The bargaining outcome implies that the firm's surplus is a constant fraction $1 - b$ of the total surplus S_t and the household's surplus is a fraction b of the total surplus.

The bargaining solution (38) and the expression for household surplus in Eq. (13) together imply that the Nash bargaining wage w_{nt}^N satisfies the Bellman equation

$$\begin{aligned} \frac{b}{1-b} (J_t^e - J_t^v) &= w_{nt}^N - \phi - \frac{\chi}{\Lambda_t} \\ &+ \mathbb{E}_t D_{t,t+1} (1 - q_{t+1}^u) (1 - \delta) \frac{b}{1-b} (J_{t+1}^e - J_{t+1}^v). \end{aligned} \quad (39)$$

In the baseline model, we assume that real wages are flexible and are given by the Nash bargaining wage (i.e., $w_{nt} = w_{nt}^N$).

2.7. Export demand

To close the model, we follow Chang et al. (2015) and specify the export demand schedule

$$X_t = \left(\tau_t \frac{P_{dt}}{P_t^*} \right)^{-\theta} X_t^* = \left(\frac{\tau_t p_{dt}}{Q_t} \right)^{-\theta} X_t^*, \quad (40)$$

where X_t^* denotes an exogenous foreign demand shifter. Demand for exported intermediate goods is inversely related to the effective price of exports, consisting of both the relative price p_{dt} , converted to foreign goods units by the real exchange rate, and the iceberg trading cost τ_t . We assume that the demand elasticity for home exports is identical to the demand elasticity for imported intermediate goods (both elasticities are given by θ).

Table 1
Calibrated parameters.

	Parameter description	Value
β	Subjective discount factor	0.99
α	Elasticity of matching function	0.50
ϕ	Unemployment benefit	0.25
b	Nash bargaining weight	0.50
δ	Job separation rate	0.10
ρ^o	Automation obsolescence rate	0.03
κ_a	Flow cost of automated production	0.98
μ	Matching efficiency	0.6606
κ	Vacancy posting per-period fixed cost	0.1128
α_n	Share of worker-produced intermediate goods	0.39
σ	Elasticity of substitution between domestic intermediate goods	2.03
$\bar{e}$	Scale of vacancy creation cost distribution	3.07
$\bar{v}$	Scale of automation cost distribution	8.57
α_d	Weight on domestic intermediate input (home bias)	0.85
θ	Substitution elasticity between domestic and imported goods	0.8
$\bar{\tau}$	Average iceberg trade cost	1.74
$\bar{Z}$	Average level of TFP	1
$\bar{s}$	Supply of skilled workers	0.3
γ_a	Share of automation equipment in production	0.32
ζ	Automation-specific productivity	3.4422
χ	Disutility of working	0.3741
ρ_z	Persistence of TFP shock	0.95
σ_z	Volatility of TFP shock	0.01
ρ_τ	Persistence of first-moment trade cost shock	0.99
σ_τ	Volatility of first-moment trade cost shock	0.00215
$\rho_{\sigma\tau}$	Persistence of trade uncertainty shock	0.96
η_τ	Volatility of trade uncertainty shock	0.37

2.8. Government policy and search equilibrium

The government finances unemployment benefit payments ϕ for unemployed workers through lump-sum taxes. We assume that the government balances the budget in each period such that

$$\phi(1 - N_t) = T_t. \quad (41)$$

In a search equilibrium, the markets for final goods, intermediate goods, and skilled labor all clear. We also assume that trade is balanced such that export revenue equals the import costs.

Market clearing for domestic intermediate goods along with that for skilled labor implies that

$$Y_{nt} = Z_t N_t, \quad Y_{at} = Z_t (\zeta A_t)^{\gamma_a} \bar{s}^{1-\gamma_a}. \quad (42)$$

Final goods market clearing requires that consumption spending, vacancy posting costs, robot operation costs, robot adoption costs, and vacancy creation costs add up to aggregate final goods output. The aggregate robot operation cost is given by $\gamma_a p_{at} Y_{at}$. Thus, the aggregate resource constraint is

$$C_t + \kappa v_t + \kappa_a \gamma_a p_{at} Y_{at} + (1 - q_{t-1}^v) v_{t-1} \int_0^{v_t^*} v dG(v) + \int_0^{J_t^v} e dF(e) = Y_t. \quad (43)$$

We focus on a balanced-trade equilibrium. In such an equilibrium, the revenue from exporting intermediate goods equals the costs of importing foreign intermediate goods, such that

$$\tau_t p_{dt} X_t = p_{ft} Y_{ft}. \quad (44)$$

We assume that the initial foreign asset holdings are $B_{-1} = 0$. Then, with balanced trade, the current account balance is also zero for all periods, and we have $B_t = 0$ for all t .

Appendix A summarizes the equilibrium conditions.

3. Parameter calibration

We use our model to study the macroeconomic impact of trade uncertainty shocks. We solve the model based on third-order approximations to the equilibrium conditions. To solve the model requires assigning values to the parameters. Table 1 shows the calibrated parameter values.

We have a quarterly model. We set the subjective discount factor to $\beta = 0.99$, such that the steady-state real interest rate is 4 percent per year. We set the matching function elasticity to $\alpha = 0.5$, in line with the literature (Blanchard and Gali, 2010; Gertler and Trigari, 2009). Following Hall and Milgrom (2008), we set the worker bargaining weight to $b = 0.5$ and the unemployment

benefit parameter to $\phi = 0.25$. Based on the data from the Job Openings and Labor Turnover Survey (JOLTS), we calibrate the steady-state job separation rate to $\delta = 0.10$ at the quarterly frequency. We set $\rho^o = 0.03$, so that automation equipment depreciates at an average annual rate of 12 percent, in line with the depreciation rate of industrial robots used by the International Federation of Robotics (IFR) for estimating the average life span of robots and for constructing their measure of the operation stocks of robots. We calibrate the vacancy posting cost $\kappa = 0.1128$ such that the flow cost of vacancy posting is about 1 percent of aggregate output. We set the matching efficiency parameter to $\mu = 0.6606$ such that the quarterly job filling rate is $q^v = 0.71$ in the steady state, as calibrated by [den Haan et al. \(2000\)](#).

We assume that the distribution functions $F(e)$ for vacancy creation costs and $G(v)$ for automation costs both follow a uniform distribution, such that

$$F(e) = \frac{e}{\bar{e}}, \quad G(v) = \frac{v}{\bar{v}}. \quad (45)$$

We calibrate the scale of the automation cost function to $\bar{v} = 8.57$ such that the model implies a steady-state automation probability of $q^a = 0.096$, or about 38 percent at the annual frequency, which lies within the range of firm-level estimates. For example, in a recent study based on the 2019 Annual Business Survey (ABS) of the U.S. Census Bureau, [Acemoglu et al. \(2022\)](#) report that, in total, 30.4 percent of U.S. workers are employed at firms using advanced technologies for automating tasks. Exposure to automation is higher in manufacturing, with 52 percent of manufacturing workers employed at firms using advanced technologies for automation. Outside of manufacturing, the exposure to automation is lower, at 28.3 percent. The model-implied automation probability in the steady state (38 percent), which corresponds to the measured automation exposure, lies within this empirical range. Furthermore, we follow [Leduc and Liu \(2024\)](#) and calibrate the scale parameter of the vacancy creation cost function to $\bar{e} = 3.07$, set the flow cost of automation to $\kappa_a = 0.98$, and calibrate the output elasticity with respect to automation equipment to $\gamma_a = 0.32$.

Based on [Firooz et al. \(2025\)](#), we calibrate the weight of worker-produced intermediate goods in final goods production to $\alpha_n = 0.39$ and the elasticity of substitution between intermediate goods produced by automation equipment and by workers to $\sigma = 2.03$.¹⁷

We normalize the average level of TFP to $\bar{Z} = 1$. We also normalize the supply of skilled workers to $\bar{s} = 0.3$, matching the median ratio of employment of college-educated workers to aggregate employment in the period from 2000 to 2019. We calibrate the average level of the automation-specific productivity to $\zeta = 3.4422$ such that the model implies a steady-state skill premium of 55 percent, in line with the observed ratio of median weekly earnings of workers with a bachelor's degree or higher to those with some college or associate degrees reported by the Bureau of Labor Statistics.

We set the average iceberg trade cost to $\bar{\tau} = 1.74$, which lies within the range of empirical estimates as surveyed by [Anderson and van Wincoop \(2004\)](#). We calibrate the weight on domestically produced intermediate goods in the aggregation technology for final goods to $\alpha_d = 0.85$, reflecting home bias in goods consumption. We calibrate the elasticity of substitution between domestic goods and imported goods to $\theta = 0.8$, which is in line with the empirical literature. For example, [Boehm et al. \(2023\)](#) find that the elasticity of trade flows to exogenous changes in tariffs is about -0.76 in the short run and about -2 in the long run (see also [di Giovanni et al., 2023](#); [Corsetti et al., 2008](#)). Since our model focuses on the short-run fluctuations induced by trade uncertainty, our calibration of $\theta = 0.8$ is consistent with the short-run elasticity estimated by [Boehm et al. \(2023\)](#). We normalize the export demand shifter to $X_t^* = 1$, which implies a steady-state export share of about 10.8 percent of GDP.

We calibrate the disutility of working to $\chi = 0.3741$ such that the model implies a steady-state unemployment rate of 5.9 percent, matching the average unemployment rate from 2000 to 2019.

For the parameters in the TFP shock processes, we set $\rho_z = 0.95$ and $\sigma_z = 0.01$, in line with the real business cycle literature. For the first-moment shock to trading costs, we set $\rho_\tau = 0.99$ and $\sigma_\tau = 0.00215$ based on the estimates of [Caldara et al. \(2020\)](#). The trade uncertainty shock parameters are also calibrated based on the study of [Caldara et al. \(2020\)](#). Specifically, we set $\rho_{\sigma\tau} = 0.96$ and $\eta_\tau = 0.37$.

4. Macroeconomic effects of trade uncertainty

To study the macroeconomic effects of trade uncertainty, we use our calibrated parameters and solve the model based on third-order perturbations around the steady-state equilibrium. We then compute impulse responses to a trade uncertainty shock following the approach of [Fernández-Villaverde et al. \(2011a\)](#).¹⁸ To illustrate the model's mechanism, we perform several counterfactual exercises.

4.1. Trade uncertainty in the baseline model

[Fig. 1](#) presents the impulse responses of several key macroeconomic variables following a one-standard-deviation shock to trade uncertainty. An increase in trade uncertainty reduces imports, redirecting production of intermediate goods from foreign sources toward domestic producers (i.e., reshoring). This expenditure-switching effect stimulates automation investment. Trade uncertainty

¹⁷ [Firooz et al. \(2025\)](#) calibrate these two parameters to target the 2016 level of robot density in the U.S. manufacturing sector of 0.02 and the cumulative increase of robot density of about 300 percent from 2002 to 2016 while the relative price of robots declined by 40 percent during the same period.

¹⁸ The impulse responses of a given variable to a trade uncertainty shock are measured by the differences between the values of that variable in the presence of the shock and its value in the stochastic steady state (i.e., its ergodic mean).

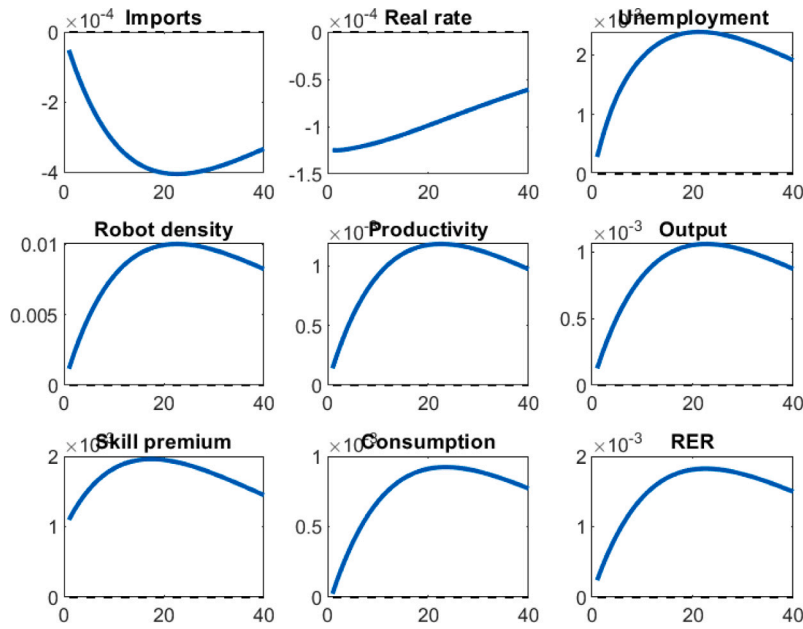


Fig. 1. Impulse responses to a trade uncertainty shock in the baseline model.

further boosts automation through a precautionary-savings channel, which lowers the real interest rate and therefore raises the present value of automation. However, trade uncertainty could discourage automation through an option-value channel. Under our calibration, the positive effects from expenditure switching and precautionary savings dominate the option-value effect, such that trade uncertainty leads to an increase in automation measured by the robot density.

Increased automation raises labor productivity, stimulating the incentive for creating new vacancies. However, with our calibration, this job-creating effect is more than offset by the job-displacing effect of automation, leading to an increase in unemployment of unskilled workers. Nonetheless, aggregate output and consumption both rise persistently because the productivity gains stemming from automation outweigh the drags from lowered imports and domestic production by unskilled workers. The automation-driven productivity gains also lowers the domestic price level, leading to a real exchange rate depreciation (i.e., an increase in Q_t).

The increased threat of automation also weakens the bargaining power of unskilled workers in wage negotiations, lowering their wages. In contrast, skilled workers are a complementary input with automation equipment. Thus, automation raises demand for skilled workers, pushing up the skilled wage while depressing the unskilled wage, resulting in a higher skill premium.

4.2. Transmission channels

The model embeds two important transmission channels for trade uncertainty shocks: an automation channel and a trade channel.

4.2.1. The automation channel

To examine the importance of the automation channel, we consider a counterfactual version of the model with a constant automation probability. In particular, we keep the automation probability q_t^a fixed at the steady-state level.

Fig. 2 shows the impulse responses to a trade uncertainty shock in the counterfactual model with a constant automation probability (red dashed line) compared to those in the baseline model (blue solid line). Absent adjustments in the automation probability, the effects of trade uncertainty on the macroeconomic variables are more muted than in the baseline model. Furthermore, in the counterfactual model, trade uncertainty reduces unemployment because it creates an expenditure switching effect, boosting demand for domestic goods. Since firms cannot adjust automation investment, they can meet the increased demand for domestic goods only by raising domestic employment. Thus, absent the automation channel, the labor-market effects of reshoring (driven by trade uncertainty) mimic a reversal of the “China shock”.

4.2.2. Trade openness

Exposure to trade (or equivalently, offshoring) is also important for the transmission of trade uncertainty shocks. To illustrate this, we consider a counterfactual with high openness to international trade. Specifically, we lower the home-bias parameter α_d to 0.6 from the baseline value of 0.85.

Fig. 3 shows the impulse responses in this counterfactual (red dashed line) versus those in the baseline model (blue solid line). When the economy is more open to international trade, the effects of trade uncertainty are amplified. Trade uncertainty leads to larger declines in imports and larger increases in robot density, unemployment, productivity, consumption, and the skill premium.

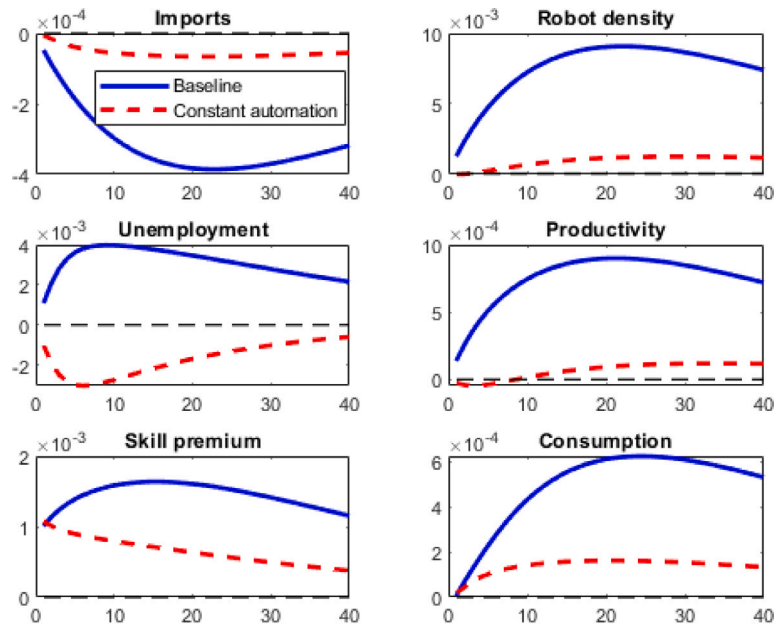


Fig. 2. Impulse responses to a trade uncertainty shock: Constant automation probability vs. the baseline model. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

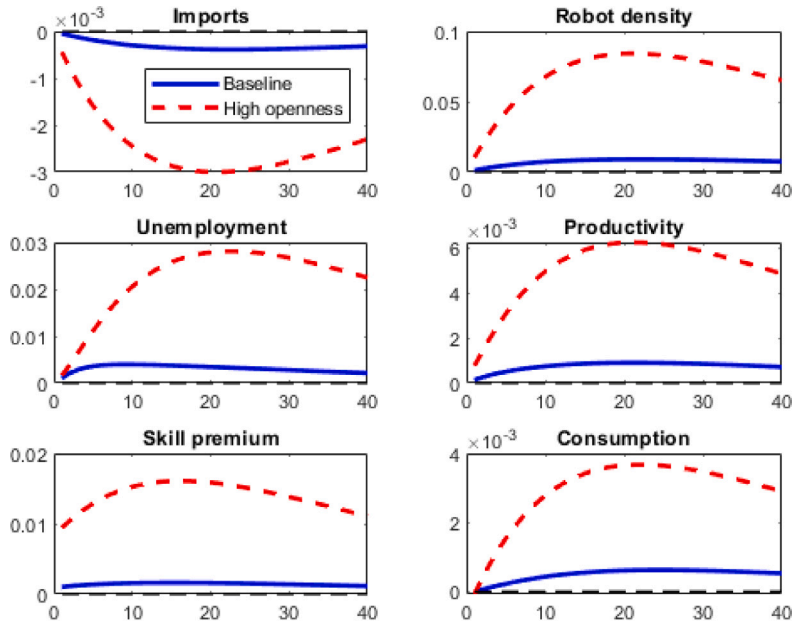


Fig. 3. Impulse responses to a trade uncertainty shock: Higher openness vs. the baseline model.

4.3. Robustness

The main results are robust to several variations of our model.

4.3.1. Capital flows

Our model features a closed capital account such that the interest rate is endogenous. As an extension, we consider an alternative framework where international capital flows are allowed. In particular, the small open economy can borrow from or lend to the rest of the world at an exogenous world interest rate r_t^* (in units of foreign consumption goods). To capture the frictions in

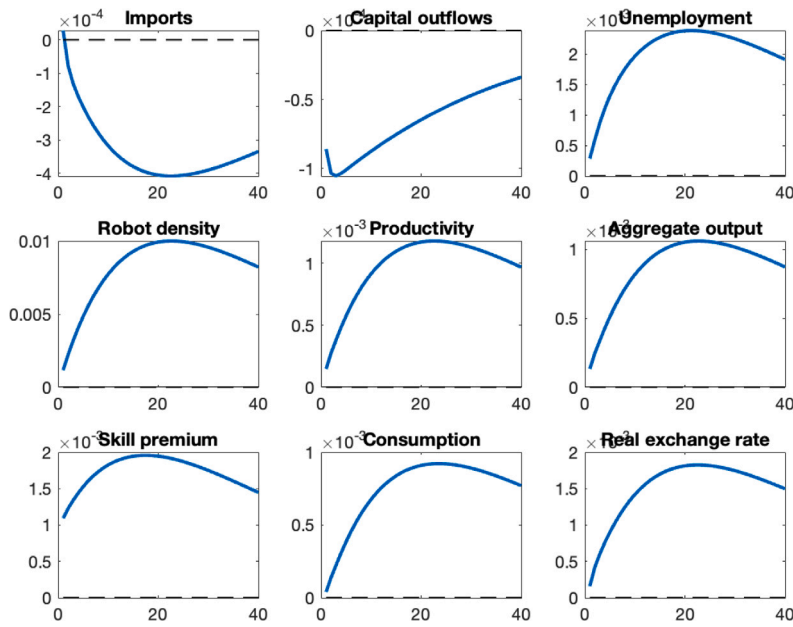


Fig. 4. Impulse responses to a trade uncertainty shock in the model with capital flows.

capital markets, we assume that changes in capital flows are subject to an adjustment cost. Since the real interest rate is fixed, the precautionary-savings channel is absent in this alternative model, and therefore, trade uncertainty boosts automation through the expenditure-switching channel only. Appendix B presents this alternative framework and equilibrium conditions.

Fig. 4 shows the impulse responses to a trade uncertainty shock in the model with capital flows.¹⁹ Increased trade uncertainty reduces imports and capital outflows, triggering an expenditure-switching effect that boosts automation. The resulting increase in robot density raises the unemployment of unskilled workers. Increased automation also boosts labor productivity, aggregate output, and the skill premium. This in turn leads to a rise in consumption and a real exchange rate depreciation. Overall, these impulse responses are qualitatively and quantitatively similar to those obtained in our baseline model under financial autarky.

4.3.2. Imported intermediate inputs for automated production

In our baseline model, a firm that operates the automation technology uses robots and skilled workers for production. However, in practice, firms' automation technology may also rely on imported intermediate goods (e.g., robot parts or other automation equipment). To incorporate this channel, Appendix C considers a generalization of the baseline model to include imported intermediate goods for automating firms. In particular, we assume that imported intermediates are a complement input to robots in the automation technology.

Fig. 5 plots the impulse responses to a trade uncertainty shock in this model. The impulse responses are similar to those in our baseline model. Trade uncertainty reduces imports and raises demand for domestic goods through an expenditure-switching effect. Since employment is a long-term relation, firms reduce hiring as the option value of waiting increases. To meet the increased demand for domestic intermediate goods, firms rely more on automation, raising the demand for robots. All else being equal, an increase in trade uncertainty would lower demand for imported equipment in the automation sector. However, the increased reliance on robots for producing domestic intermediate goods raises the demand for imported equipment since imported intermediate inputs complement robots. Overall, trade uncertainty reduces imported goods and domestic employment, and raises robot density and imported equipment for automated production. As in the baseline model, the increased automation also boosts labor productivity and the skill premium.

4.3.3. Persistence of trade uncertainty

Trade uncertainty may be more persistent than past data suggest for the calibration of the baseline model. Trade tensions, geopolitical conflicts, and climate change risks may be part of a new normal with persistently elevated trade uncertainty. We consider a counterfactual case with a higher persistence of the trade uncertainty shock by raising the persistence parameter $\rho_{\sigma\tau}$ from 0.96 in the baseline calibration to 0.99, proxying for a quasi-permanent regime with higher trade uncertainty.

Fig. 6 shows the impulse responses in this counterfactual case (red dashed line) versus those in the baseline model (blue solid line). Near-permanent trade uncertainty generates a stronger expenditure-switching effect, resulting in greater reshoring (i.e., larger

¹⁹ We calibrate the bond adjustment cost parameter to $\psi = 2$ for solving the model.

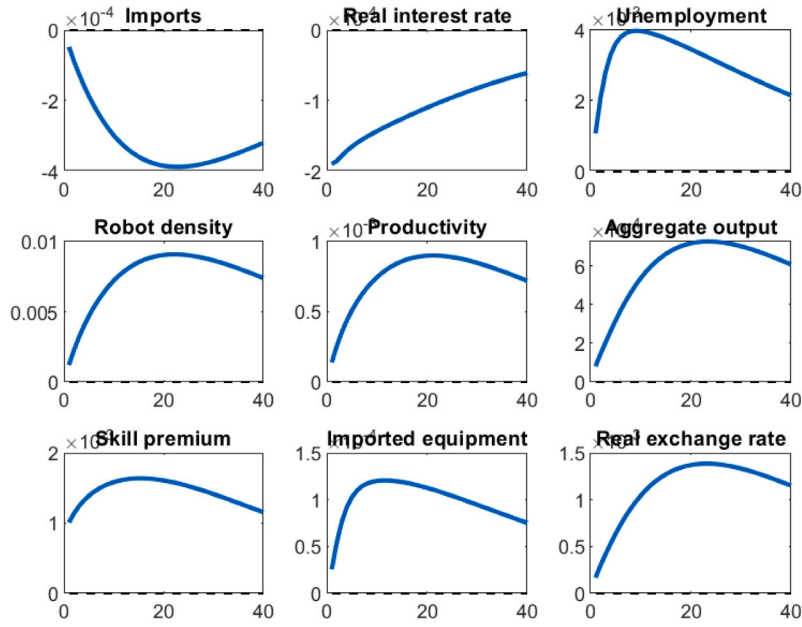


Fig. 5. Impulse responses to a trade uncertainty shock in the model with imported equipment for automated production.

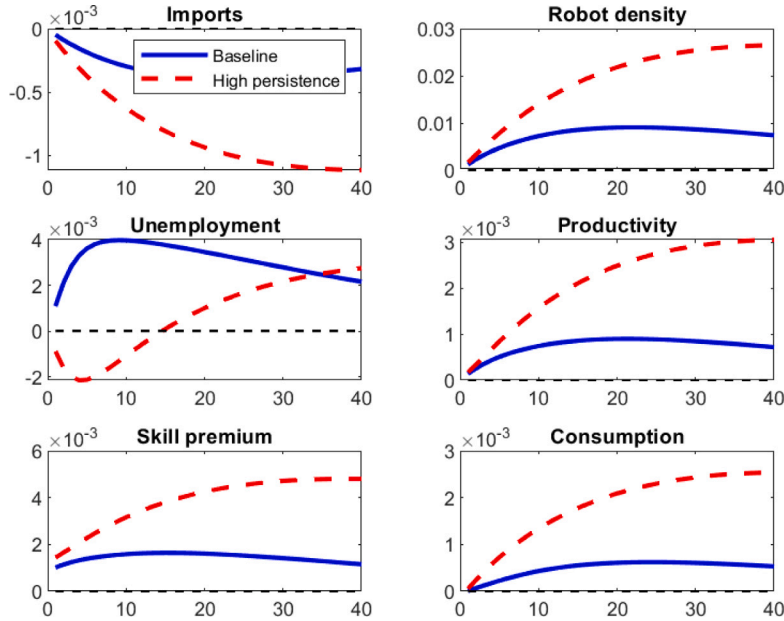


Fig. 6. Impulse responses to a trade uncertainty shock: More persistent trade uncertainty shock vs. the baseline model.

declines in imports) and a larger increase in automation investment. The stronger expenditure-switching effect in this case is such that it raises domestic employment of unskilled workers in the short run, although the job displacing effects of automation dominates over time, leading to a rise in unemployment. The larger boom in automation investment also results in greater gains in productivity and larger increases in the skill premium and consumption than in the baseline model.

4.3.4. The role of wage rigidity

In the baseline model, we assume that real wages are flexible. We now examine the robustness of the results to wage stickiness. Following the literature (Hall, 2005; Shimer, 2005), we assume that the real wage of unskilled workers is a geometrically weighted average of the Nash bargaining wage and the wage rate in the previous period, such that

$$w_{nt} = w_{n,t-1}^{\gamma_w} (w_{nt}^N)^{1-\gamma_w}, \quad (46)$$

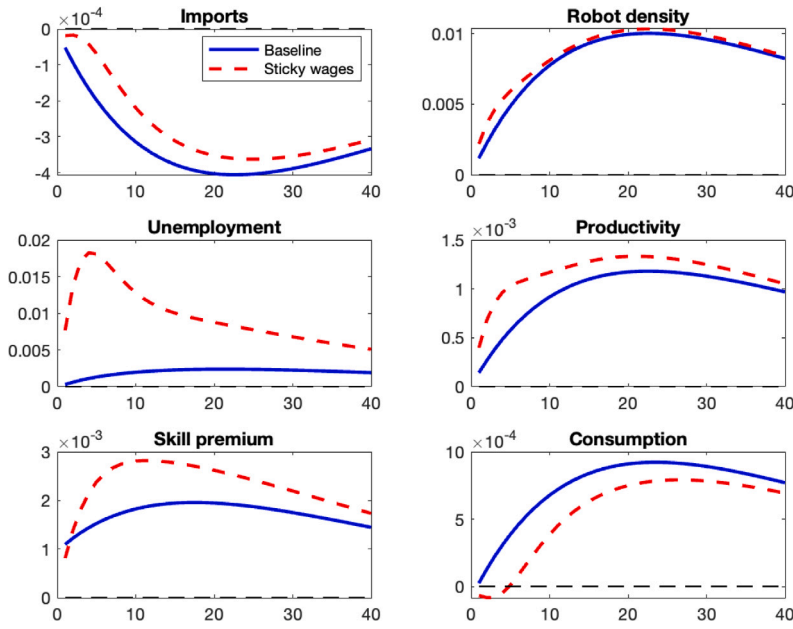


Fig. 7. Impulse responses to a trade uncertainty shock: Sticky wages vs. the baseline model.

where $\gamma_w \in (0, 1)$ represents the degree of real wage rigidity. We follow [Leduc and Liu \(2016\)](#) and set the real wage rigidity parameter to $\gamma_w = 0.8$, which is in line with [Gertler and Trigari \(2009\)](#), who calibrate the probability of nonrenegotiation of wage contracts at 0.89.

[Fig. 7](#) compares the impulse responses from the case with wage rigidities (red dashed line) to those in the baseline case with flexible wages (blue solid line). As in the standard DMP framework, wage rigidities amplify the increase in unemployment following the trade uncertainty shock, reflecting the Shimer volatility puzzle ([Shimer, 2005](#); [Hall, 2005](#)). The impulse responses of the other macroeconomic variables are similar to those in the baseline model.

4.3.5. Delivery lags

In the baseline model, we assume that importing intermediate inputs for final goods production requires a delivery lag. Appendix D.1 presents a version of the model without delivery lags. The impulse responses to trade uncertainty in the model without delivery lags are similar to those obtained in the baseline model. In particular, trade uncertainty reduces imports, and increases unemployment and robot density. We note that trade uncertainty in this version of the model still activates the expenditure-switching channel because of the curvature in the import demand schedule.

4.3.6. Tariff uncertainty

Our benchmark model includes iceberg trade costs. Appendix D.2 presents a model with tariffs on imported intermediate goods, instead of iceberg costs. We show that the effects of a second-moment shock to tariffs are similar to those of a second-moment shock to the iceberg costs, although the magnitude of the responses is smaller.

4.3.7. Other shocks

Appendix D.3 presents the macroeconomic effects of other shocks, including first-moment shocks to trade costs and both first- and second-moment shocks to TFP. The impulse responses to these shocks are quite different from those to trade uncertainty.

4.4. Welfare consequences of automation

The labor search frictions in our model lead to a congestion externality as in the standard DMP model, implying that the competitive equilibrium allocations are not necessarily Pareto optimal. More importantly, fluctuations in the automation probability lead to endogenous fluctuations in the workers' relative bargaining power in wage negotiations ([Leduc and Liu, 2024](#)). Thus, even if the Hosios condition holds (i.e., the bargaining weight b is equal to the elasticity of the matching function α , which is true under our calibration), endogenous variations in the worker bargaining power driven by fluctuations in automation could push the equilibrium allocations away from the social optimum.

To examine the welfare consequences of endogenous automation, we compute the welfare gains of moving from our baseline model to a counterfactual economy with a constant automation probability (i.e., the counterfactual model that we have examined in Section 4.2.1). Under our calibrated parameters, allowing automation to fluctuate (as in our baseline model) incurs a modest welfare

loss of about 0.75 percent of consumption equivalent.²⁰ This result is driven by the fact that automation amplifies macroeconomic fluctuations as shown in Fig. 2 and thus reduces the welfare of the risk-averse representative household.²¹

5. Empirical evidence

Our theoretical model predicts that trade uncertainty can stimulate automation investment and reduce imported intermediate goods. The increased automation driven by trade uncertainty in turn raises labor productivity, value added, and the skill premium and reduces domestic employment. We now present some empirical evidence supporting the model predictions.

5.1. Data

We measure trade uncertainty using the U.S. TPU index constructed by Caldara et al. (2020), which is based on the frequency of articles in several major U.S. newspapers that discuss economic policy uncertainty and contain one or more phrases related to trade policy (such as “import tariffs”, “import barriers”, “WTO”, “trade policy”, and “trade agreement”). The monthly TPU index is available starting from 1960.²²

We measure automation using robot density in U.S. industries. Specifically, we define robot density in industry j and year t ($Robot_{jt}$) as the operational stock of industrial robots per thousand employees. We obtain data on industrial robots for each two-digit ISIC industry from the International Federation of Robotics (IFR). We obtain employment data for three-digit North American Industry Classification System (NAICS 2017) industries from the Bureau of Labor Statistics (BLS) and cross-walk these industries to ISIC codes. The matched sample contains 14 industries (at the ISIC two-digit level) for the years 2004 to 2022.

To help explore the differential effects of trade uncertainty across industries with different exposure to offshoring, we construct a measure of industry-level offshoring exposure using the initial share of imported intermediate goods in gross output (i.e., in the beginning year of our sample) for two-digit ISIC industries. We obtain data on the gross imports of intermediate products from OECD Trade in Value-Added, and on gross output from the Bureau of Economic Analysis (BEA). For each industry-year pair, we also use the import-weighted average of tariffs that the U.S. imposes on its imports from the World Integrated Trade Solution (WITS). The annual sample of imports covers 14 two-digit ISIC industries for the years from 1997 to 2020.

We measure labor productivity for a two-digit ISIC industry by the ratio of real value added to total employment in that industry, with value-added data sourced from the BEA. We construct a measure of the skill premium using data from the Current Population Survey (CPS). In particular, the skill premium is measured by the earnings per hour of skilled workers (i.e., with a college degree or above) divided by those of unskilled workers (with a high school degree). The annual sample covers the period from 1997 to 2022.

Since we have annual data on industrial robots and imports of intermediate goods, we aggregate the TPU index from a monthly frequency to an annual frequency by taking the within-year average.

Table 2 reports the summary statistics of our data. Robot density (in log units) in the data displays substantial variations across industries and time, with a standard deviation of 2.91, which is over 10 times its sample mean (in absolute value). The interaction between TPU (in log units) and the initial share of intermediate imports also exhibits significant variations, with a standard deviation of about 90 percent of its mean. The share of imported intermediate goods (in log units) – our measure of offshoring activity – has more modest variations across industries and over time, with a standard deviation of about 60 percent of its mean (in absolute value). The real outcome variables, including labor productivity, employment, value-added, and the skill premium, are relatively stable, with standard deviations between 15 and 50 percent of their respective means.

5.2. Trade uncertainty, automation, and offshoring

To examine the empirical relationship between automation and offshoring with trade uncertainty, we consider the following empirical specification

$$\ln Robot_{jt} = \alpha_0 + \alpha_1 ImpShare_j \times \ln TPU_t + \alpha_2 \ln(1 + Tariff_{jt}) + \eta_j + \theta_t + \varepsilon_{jt}, \quad (48)$$

where $ImpShare_j$ is the share of imported intermediate goods in gross output for industry j at the beginning of our sample (2004), as a proxy for the initial exposure of the industry to offshoring. The terms η_j and θ_t denote industry and time fixed effects, respectively, and ε_{jt} denotes the regression residuals. In the regression, we include industry-level tariffs (i.e., $\ln(1 + Tariff_{jt})$) as an additional control variable to mitigate potential confounding effects of changes in trade barriers.

²⁰ The welfare gain (or loss) of keeping the automation probability constant relative to the baseline economy is measured by a constant Δ such that

$$V_t = \sum_{i=0}^{\infty} \beta^i [\log(\tilde{C}_t(1 + \Delta)) - \chi \tilde{N}_t] = \frac{\log(1 + \Delta)}{1 - \beta} + \tilde{V}_t, \quad (49)$$

where V_t and $\tilde{V}_t$ denote, respectively, the welfare in the baseline economy and the counterfactual and $\tilde{C}_t$ and $\tilde{N}_t$ denote the consumption and employment in the counterfactual.

²¹ Our framework can be extended to study the welfare implications of tax policies in a model with automation and labor search frictions, which is an important subject for future research.

²² Caldara et al. (2020) also develop a firm-level measure of TPU and another aggregate measure of TPU based on a stochastic volatility model for U.S. import tariffs.

Table 2
Summary statistics.

	Mean	Count	SD	Min	Max	IQR
log Robot density	-.222	258	2.906	-7.079	5.027	4.341
log $TPU \times$						
Initial share of intermediate imports	.428	390	.373	.002	1.977	.275
log Share of intermediate imports	-2.414	360	1.514	-7.695	-.445	1.049
log Labor Productivity	-1.957	364	.999	-3.734	.823	1.358
log Employment	6.778	364	.998	4.168	8.572	1.357
log Real Value-Added	4.947	390	.851	3.211	6.839	1.199
log Skill premium	.464	390	.105	.192	.768	.157
log(1+Tariff)	.018	364	.022	0	.114	.014

Note: The table shows the summary statistics of the variables used in the regressions. Robot density is defined as the operation stock of industrial robots per thousand employees in each industry. The share of intermediate imports is the ratio of imported intermediate goods to gross output in each industry. TPU is the trade policy uncertainty index, which is an aggregate time series constructed by [Caldara et al. \(2020\)](#). Labor productivity is the ratio of value-added to employment in each industry. Skill premium is the ratio of hourly earnings of workers with a college degree or above to those with a high school education. Tariff is the industry-year-specific import-weighted average of tariffs that the U.S. imposes on its imports. See the text for data sources.

Table 3
Trade policy uncertainty, automation, and offshoring.

	(1) log(Robot density)	(2) log(Import share)
Initial import share $\times$ log(TPU)	2.857* (1.464)	-0.941** (0.396)
log(1+Tariff)	34.99 (59.09)	-7.344 (8.649)
Industry fixed effect	✓	✓
Time fixed effect	✓	✓
Observations	241	336
R ²	0.923	0.956
Years	2004:2022	1997:2020
No. of industries	14	14

Note: Column (1) reports the estimates of the regression of robot density on trade uncertainty proxied by the interaction between TPU and initial exposure to offshoring. Column (2) reports the estimates of the regression of the share of imported intermediate goods in gross output on trade uncertainty. Both regressions control for industry and time fixed effects as well as industry-time-specific tariffs that the U.S. imposes on its imports. Standard errors clustered at the industry level are shown in parentheses. The levels of statistical significance are denoted by asterisks: *** for $p < 0.01$, ** for $p < 0.05$, and * for $p < 0.1$.

The key parameter of interest is α_1 , which measures the sensitivity of an industry's robot density to changes in trade policy uncertainty, depending on the industry's initial exposure to offshoring. In what follows, we refer to the interaction between TPU (in log units) and the import share as "trade uncertainty exposures". Our theory suggests that an increase in trade uncertainty should be associated with an increase in robot density, and this response should be stronger for industries that are more exposed to offshoring. Specifically, the impulse responses in [Fig. 3](#) show that, in a more open economy, trade uncertainty should lead to a larger increase in robot density. Thus, the theory predicts that $\alpha_1 > 0$.

This prediction is supported by the data, as shown in [Table 3](#) (Column (1)). The table shows that, after controlling for the industry and time fixed effects as well as tariffs, an increase in TPU is associated with a larger increase in robot density in industries that are more exposed to offshoring. This correlation is statistically significant at the 90 percent confidence level and economically important. A one-standard-deviation increase in trade uncertainty exposures is associated with an increase in robot density of about 1.07 log points ($2.857 \times 0.373 \approx 1.07$), which is about a third of the standard deviation of the logarithm of the robot density (2.91).

Our model also predicts that heightened trade uncertainty reduces offshoring, especially in industries that are initially more exposed to offshoring (see the impulse response of imports to trade uncertainty in [Fig. 3](#)). This model prediction aligns well with the empirical evidence. [Table 3](#) (Column (2)) shows that, controlling for industry and time fixed effects as well as industry tariffs, an increase in TPU is associated with a larger decline in offshoring for industries that are initially more exposed to importing of intermediate goods.²³ The estimated negative correlation between trade uncertainty and imports of intermediate goods is also economically meaningful. A one-standard-deviation increase in trade uncertainty exposures is associated with a reduction in the share of imported intermediate goods of about 0.35 log points ($-0.941 \times 0.373 \approx -0.35$), which is almost a quarter of the standard deviation of the import share.

²³ We show in the Appendix that TPU has heterogeneous effects on imports from different origin countries. In particular, Table E.1 shows that an increase in TPU is associated with a large and significant decline in imports from China, partly reflecting the effects of sharp increases in bilateral trade tensions (beyond that explained by tariffs) between the United States and China.

Table 4
Trade policy uncertainty, offshoring, and macroeconomic variables.

	(1) log(Labor productivity)	(2) log(Employment)	(3) log(Value Added)	(4) log(Skill premium)
Initial import share $\times$ log(TPU)	0.876*** (0.268)	−0.009 (0.215)	0.867*** (0.202)	0.102 (0.0648)
log(1+Tariff)	−0.087 (5.528)	−0.208 (9.788)	−0.295 (7.924)	1.915 (1.696)
Industry fixed effect	✓	✓	✓	✓
Time fixed effect	✓	✓	✓	✓
Observations	364	364	364	364
R ²	0.971	0.979	0.944	0.767
Years	1997:2022	1997:2022	1997:2022	1997:2022
No. of industries	14	14	14	14

Note: Columns (1), (2), (3), and (4) report the results of regressing labor productivity, employment, value-added, and skill premium, respectively, on the interaction between TPU and initial exposure to offshoring. All regressions control for industry and time fixed effects as well as industry-level tariffs imposed by the U.S. Standard errors clustered at the industry level are shown in parentheses. The levels of statistical significance are denoted by asterisks: *** for $p < 0.01$, ** for $p < 0.05$, and * for $p < 0.1$.

5.3. Trade uncertainty and other macroeconomic variables

Our model further predicts that heightened trade policy uncertainty should increase labor productivity, the skill premium, and value added, while reducing employment (see Fig. 1). These model predictions are broadly consistent with the data, as shown in Table 4.

The table shows the same regressions as in Eq. (48), where we replace the dependent variable with each of the macroeconomic variables of interest. As shown in the table, an increase in TPU is associated with a greater increase in labor productivity and value added in industries more exposed to offshoring in the initial period. These effects are statistically significant and economically important. In particular, a one-standard-deviation increase in trade uncertainty exposures is associated with an increase in labor productivity of about 0.33 log points ($0.876 \times 0.373 \approx 0.33$), which is about a third of the standard deviation of labor productivity. The same increase in trade uncertainty exposures is associated with an increase in value added of about 0.32 log points ($0.867 \times 0.373 \approx 0.32$), which is about 38 percent of the standard deviation of value added.

The correlation between TPU and employment and that between TPU and skill premium are imprecisely estimated, reflecting the noise in the relatively small sample. However, the sign of the estimated coefficients are in line with our theoretical predictions.

5.4. The automation channel

In our model, the effects of trade uncertainty on employment, labor productivity, output, and the skill premium work through the automation channel. Specifically, as shown in Fig. 2, the automation channel amplifies the responses of labor productivity and the skill premium to a trade uncertainty shock. Trade uncertainty also reduces low-skilled employment in our baseline model, whereas it raises employment in the counterfactual economy with a constant automation probability.

We now present some empirical evidence that is consistent with our model's automation channel. Trade uncertainty can influence macroeconomic variables through multiple channels. To highlight the automation channel, we follow the two-stage estimation procedure of Bertrand and Mullainathan (2001). In the first stage, we regress robot density on the interactions of TPU with initial exposure to offshoring, controlling for industry and time fixed effects. In the second stage, we regress the variables of interest (labor productivity, skill premium, etc.) on the predicted robot density from the first-stage regression, controlling for industry tariffs. We interpret the estimated coefficient on the predicted robot density in the second-stage regression (shown in Table 5) as reflecting the sensitivity of those macroeconomic variables to trade policy uncertainty through the automation channel.

Table 5 shows that an increase in robot density driven by an increase in trade uncertainty is associated with a statistically significant increase in labor productivity, value added, and skill premium. An increase in robot density driven by trade uncertainty also reduces employment, although it is not statistically significant.

The responses of labor productivity, value added, and the skill premium to trade uncertainty through the automation channel are economically important. A one-standard-deviation increase in trade uncertainty exposures is associated with an increase in robot density of 1.07 log points (as shown in the first-stage regression). Working through this automation channel, trade uncertainty raises labor productivity by about 30 percent of its standard deviation ($1.07 \times 0.279/0.999 \approx 0.3$). Trade uncertainty also raises value added by about 21 percent, or a quarter of its standard deviation ($1.07 \times 0.2/0.851 \approx 0.25$) and it increases the skill premium by about 6 percent, which is 58 percent of its standard deviation ($1.07 \times 0.057/0.105 \approx 0.58$). These results suggest that the automation channel is empirically important for the transmission of trade policy uncertainty.

Table 5
Two-stage least squares: Empirical importance of automation.

	(1) log(Labor productivity)	(2) log(Employment)	(3) log(Value Added)	(4) log(Skill premium)
Predicted log(Robot density)	0.279** (0.119)	−0.0799 (0.0541)	0.200** (0.0961)	0.0568** (0.0289)
log(1+Tariff)	−13.72 (10.86)	0.411 (4.831)	−13.31 (10.60)	2.493 (4.200)
Industry fixed effect	✓	✓	✓	✓
Time fixed effect	✓	✓	✓	✓
Observations	241	241	241	241
Years	2004:2022	2004:2022	2004:2022	2004:2022
No. of industries	14	14	14	14

Note: This table shows the second-stage regressions using the robot density predicted from the first-stage regression shown in Column (1) of Table 3 as the regressor. All regressions control for industry and time fixed effects as well as industry-time-specific tariffs that the U.S. imposes on its imports. Standard errors clustered at the industry level are shown in parentheses. The levels of statistical significance are denoted by asterisks: *** for $p < 0.01$, ** for $p < 0.05$, and * for $p < 0.1$.

6. Conclusion

Trade uncertainty has risen in recent years, stemming from risks associated with tariffs, geopolitical tensions, and climate change. This uncertainty has led to a reconsideration of the costs and benefits of offshoring to lower production costs.

In this paper, we have examined how automation affects domestic labor markets when trade uncertainty creates the incentive to reshore production processes from foreign sources back to the domestic market. In our model, domestic firms can produce intermediate goods using either a labor-only technology or an automation technology. Through an expenditure-switching effect, heightened trade uncertainty raises domestic production but not necessarily domestic employment because automation is a labor-substituting technology. Although automation raises productivity and thus labor demand, the job-displacing effect dominates under our calibration. As such, trade uncertainty boosts automation investment while raising unemployment of unskilled workers. Increased automation also leads to a higher skill premium.

Our model's predictions are in line with industry-level empirical evidence. Our evidence suggests that, in industries more exposed to offshoring, heightened trade uncertainty reduces offshoring while stimulating automation relative to other industries. Consistent with our model's predictions, this translates into higher productivity and pushes up the skill premium while lowering employment.

We focus on the positive aspects of the interactions between reshoring, automation, employment, and wages, taking government policy as given. Our model implies that the threat of automation (e.g., stemming from trade uncertainty) could weaken the bargaining power of unskilled workers. Such endogenous variations in workers' bargaining power can create a potential source of inefficiency that may call for policy interventions. Studying policy implications in a theoretical framework like ours is a promising avenue for future research, and it would complement the recent work of Grossman et al. (2023), who examine optimal policy in a model with critical production input in the face of global supply chain disruptions.

Declaration of competing interest

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.jinteco.2025.104091>.

Data availability

Replication package for "Reshoring, Automation, and Labor Markets under Trade Uncertainty" (Original data) (Mendeley Data)

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